

Riccardo Orlando

PH.D. IN DEEP LEARNING AND NATURAL LANGUAGE PROCESSING

☎ (+39) 3470089525 | ✉ orlandoricc@gmail.com | 🏠 riccardorlando.xyz | 📷 riccorl | 🌐 riccardo-orlando | 🎓 Riccardo Orlando

Experience

AI Scientist

Rome, Italy

WORLD BANK GROUP

Mar. 2025 - Current

- Drove the research-to-production lifecycle for ImpactAI's core agentic pipeline, designing and deploying agent orchestration, evidence retrieval, and Entity Linking systems to automate complex meta-analyses.
- Leading the data generation and annotation effort to create a benchmark for retrieval systems over Randomized Controlled Trials (RCTs), enabling rigorous evaluation of RAG performance on scientific literature.
- Led end-to-end development of a multi-source information extraction pipeline (event extraction, entity linking, relation extraction), achieving 20–30% improvement over baseline by fine-tuning decoder-based LLMs for high-accuracy inference and encoder models for low-latency serving, supported by synthetic data generation pipelines and custom benchmark design.

Researcher

Rome, Italy

SAPIENZA UNIVERSITY OF ROME

Nov. 2021 - Jan. 2025

- Led model pretraining and data preprocessing for Minerva (1B, 3B, and 7B) LLMs, the first family of Open Foundation Models trained from scratch on Italian data.
- Developed ReLiK, a novel retrieval-augmented system achieving SOTA performance in Entity Linking and Relation Extraction.
- Led the data annotation process that resulted in a novel dataset for Semantic Role Labeling.
- Developed production-ready pipelines for several state-of-the-art NLU tasks.
- Served as Teaching Assistant for MSc courses in Natural Language Processing.

NLP Engineer

Rome, Italy

BABELSCAPE

Feb. 2021 - Nov. 2021

- Designed and implemented end-to-end multilingual NLP pipelines for production deployment.
- Built scalable ML serving framework that drastically reduced model deployment time.
- Authored and presented two papers at EMNLP 2021.

Software Engineer

Rome, Italy

CAPGEMINI

Mar. 2018 - Sept. 2018

- Developed a back-end application using Spring Boot to handle internal workflows for different customers.
- Contributed to front-end development using Angular.

Education

Ph.D. in Computer Science (Deep Learning & Natural Language Processing)

Rome, Italy

SAPIENZA UNIVERSITY OF ROME

Nov. 2021 - Jan. 2025

- Thesis: Enhancing Semantic Understanding Across Multiple Dimensions: Towards A Unified Framework for Semantic Knowledge Extraction.
- Advisor: Roberto Navigli

Master of Science in Engineering in Computer Science

Rome, Italy

SAPIENZA UNIVERSITY OF ROME

Oct. 2018 - Jan. 2021

- Final Grade: 110/110 with honors
- Thesis: Automatic Approaches to Produce Multilingual Training Data for Semantic Role Labelling.

Bachelor of Science in Computer Science

Rome, Italy

SAPIENZA UNIVERSITY OF ROME

Oct. 2014 - May 2018

- Final Grade: 101/110
- Thesis: Densest Subgraph in Fork/Join.

Relevant Publications

- | | | |
|------|--|--------------|
| 2025 | Karim Ghonim, Pere-Lluís Huguet Cabot, <u>Riccardo Orlando</u> , and Roberto Navigli. RAED: Retrieval-Augmented Entity Description Generation for Emerging Entity Linking and Disambiguation. | EMNLP 2025 |
| 2024 | Francesco Maria Molfese, Simone Conia, <u>Riccardo Orlando</u> , and Roberto Navigli. ZEBRA: Zero-Shot Example-Based Retrieval Augmentation for Commonsense Question Answering. | EMNLP 2024 |
| 2024 | <u>Riccardo Orlando</u> , Luca Moroni, Pere-Lluís Huguet Cabot, Edoardo Barba, Simone Conia, Sergio Orlandini, Giuseppe Fiameni, and Roberto Navigli. Minerva LLMs: The First Family of Large Language Models Trained from Scratch on Italian Data. | CLiC-it 2024 |

2024	Riccardo Orlando , Pere-Lluís Huguet Cabot, Edoardo Barba, and Roberto Navigli. ReLiK: Retrieve and LinK, Fast and Accurate Entity Linking and Relation Extraction on an Academic Budget.	ACL 2024
2024	Simone Conia, Edoardo Barba, Abelardo Carlos Martínez Lorenzo, Pere-Lluís Huguet Cabot, Riccardo Orlando , Luigi Procopio, Roberto Navigli. MOSAICo: a Multilingual Open-text Semantically Annotated Interlinked Corpus.	NAACL 2024
2023	Riccardo Orlando , Simone Conia, and Roberto Navigli. Exploring Non-Verbal Predicates in Semantic Role Labeling: Challenges and Opportunities.	ACL 2023
2022	Riccardo Orlando , Simone Conia, Stefano Faralli, and Roberto Navigli. Universal Semantic Annotator: the First Unified API for WSD, SRL and Semantic Parsing.	LREC 2022
2021	Riccardo Orlando , Simone Conia, Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. AMuSE-WSD: An All-in-one Multilingual System for Easy Word Sense Disambiguation.	EMNLP 2021
2021	Simone Conia, Riccardo Orlando , Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. InVeRo-XL: Making Cross-Lingual Semantic Role Labeling Accessible with Intelligent Verbs and Roles.	EMNLP 2021

Skills

Frameworks	Deep Learning & Machine Learning (PyTorch, PyTorch Lightning, Hydra, Transformers, ONNX, TensorFlow, scikit-learn)
Back-end	FastAPI, Django, REST API
Languages	Python, Java, Kotlin, Bash/Shell
Developer Tools	Git, Docker, VS Code, Visual Studio, PyCharm, IntelliJ

Honors & Awards

2024	Winner , 3x Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type C for compute hours at CINECA, the Italian HPC Center	Rome, Italy
2023	Winner , Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type B for compute hours at CINECA, the Italian HPC Center	Rome, Italy
2022	Winner , Call for Proposals for Scientific Research	Rome, Italy

Program Committees

2025	Reviewer , ARR May 2025
2024	Reviewer , ARR February 2024, ARR June 2024
2023	Reviewer , ARR December 2023, ARR October 2023, EMNLP 2023, ACL 2023, EACL 2023
2022	Reviewer , LREC 2022
2021	Reviewer , EMNLP 2021